



Indian Association for the Cultivation of Science  
(Deemed to be University under the *de novo* category)

Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's  
Program/PhD Course

Mid-Semester Examination-Autumn 2025

Subject: Introductory Mathematical Methods  
and Classical Mechanics

Subject Code(s): PHS1101

Full marks: 25

Time allotted: 2 hr

Answer all questions

1. (a) Consider a surface  $\phi(x, y, z) = x^2y + y^2z + z^2x = \text{constant}$ . Find a unit vector, perpendicular to this surface at the point  $(1, 1, 1)$ . (3 marks)  
(b) Prove that  $\nabla \cdot (\nabla \times \vec{A}) = 0$ , for an arbitrary vector field  $\vec{A}(x, y, z)$ . (2 marks)
2. (a) Verify Gauss's theorem for the vector field  $\vec{A} = xy\hat{i} + yz\hat{j} + zx\hat{k}$ , considering a unit cube with the origin  $(0, 0, 0)$  at one of its corner. (3 marks)  
(b) Show that  $\nabla^2(1/r^2) = 0$ , provided  $r \neq 0$ . (2 marks)
3. (a) Consider a damped oscillator of mass  $m$ , with restoring force constant  $k$  and resistive force constant  $k_1$ . If the initial conditions are  $x = 0$  and  $\dot{x} = v_0$  at  $t = 0$ , find its displacement  $x$  at anytime  $t$ . (3 marks)  
(b) In the above problem, when will the motion be critically damped, and when will it become over damped? Plot  $x$  versus  $t$  in these two cases. (2 marks)
4. (a) Find the radial and cross-radial acceleration for a motion in a plane. (3 marks)  
(b) Show that for a central force, torque is zero and the angular momentum is conserved about the centre of the force. Also show that angular momentum conservation leads to constant rate of sweeping of the area by the radius vector. (2 marks)
5. (a) Show that for a central force,  $\vec{F} = f(r)\hat{r}$ ,  $\nabla \times \vec{F} = 0$  ( $\hat{r}$  is the unit vector along  $r$ ) (2 marks)  
(b) From  $\nabla \times \vec{F} = 0$ , using Stokes theorem, prove that the work done by the force is independent of the path. From the above result, define the concept of potential of  $\vec{F}$  as,  $\vec{F} = -\nabla V$ . Show that the sum of kinetic and potential energy for a motion in such a force field is conserved. (3 marks)